

Devesh Maheshwari

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EDUCATION

Purdue University – West Lafayette

Bachelor of Science – Electrical Engineering & Computer Science

May 2027

GPA: 3.98 / 4.00

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Go, Java, SQL, JavaScript/TypeScript, Assembly x86-64

Frameworks & AI/ML: React, Node.js, REST APIs, LLM APIs, PyTorch / TensorFlow, scikit-learn, RAG (FAISS)

Embedded Systems: Embedded C, STM32, RP2350B/Raspberry Pi, Arduino, Verilog, VHDL, RTOS

Protocols & Interfaces: I²C, SPI, UART, USB, PWM, ADC, Digital Signal Processing

HW + Simulation: Oscilloscopes, multimeters, logic analyzer, soldering, LTspice, KiCad, MATLAB/Simulink, Synopsys TCAD

Tools & Infra: Linux, Git, GitHub, Jira, AWS, S3, SQL, MongoDB, GitLab, Docker, CI/CD, Kubernetes, Postgres, Flink

EXPERIENCE

Palo Alto Networks – Software Engineering Intern

Santa Clara, CA | May 2026 – Aug 2026

- Building CDC pipelines (Debezium, Kafka, Flink SQL) to replicate Postgres microservices data into analytics Aurora DB.
- Enabling zero-downtime schema migrations via Goose view-aliasing, with Postgres row-level security for tenant isolation.
- Deploying pipelines on Kubernetes (EKS) with Helm; a Go metrics service exports aggregates to S3 for Tableau dashboards.
- Developing Claude agent automating metric onboarding, collapsing multi-day, 4-branch workflow into one invocation.

StudioX-AI – AI / ML Intern

Remote, San Jose, CA | Jun 2025 – Aug 2025

- Automated DevOps and customer support with AI workflows, boosting internal efficiency and saving 5-8 hrs weekly.
- Integrated GitLab, Docker APIs with Claude via YAML for CI/CD and user management, sanity checks, and image builds.
- Packed integrations into DevOps chatbot enabling 10+ operations through conversational interface, reducing manual effort.
- Streamlined customer email intake with Claude, semantically analyzing ~50 emails/month to generate tickets.

Sopra Steria – Applied AI Intern

Noida, India | May 2024 – Jul 2024

- Delivered RAG search over employee PDFs, reducing lookup time and generating grounded answers in under 5s.
- Synthesized ingestion pipeline by parsing PDFs, chunking ~800 tokens, embedding with OpenAI, indexing in FAISS db.
- Connected FAISS retrieval with GPT-4 to synthesize answers from top passages, improving response accuracy to 90%.

Bytedge – AI Developer

New Delhi, India | May 2023 – Jul 2023

- Enabled natural language search on company platform, cutting lookup from minutes to under 10s for 200+ daily queries.
- Fine-tuned GPT-3 with ~300 prompts to generate SQL tailored to company schema, retrieving records from complex menus.
- Implemented GPT-3 outputs with summarization pipeline, presenting results in natural language within search bar interface.

PROJECTS

Wireless Laptop Motion-Controller – On-Device ML

June 2026 – Present

- Building bluetooth enabled wrist motion controller using RP2350B MCU and 6-axis IMU to classify hand motions on-device.
- Training custom 1D CNN (Python/TensorFlow-Keras) on raw IMU windows, quantizing to int8 for TFLite Micro inference.
- Developing real-time gesture-spotting firmware in C, BLE pipeline to host app driving OS-level media/volume/scroll controls

Gesture-Driven Game Console – Raspberry Pi Pico

Mar 2026 – May 2026

- Built Frogger-style arcade game on an RP2350, played via real-time hand swipes read by APDS-9960 gesture sensor over I²C.
- Ran gesture detection and the game's state machine on separate RP2350 cores, linked through the inter-core hardware FIFO.
- Rendered graphics to SPI LCD with PWM sound effects, hardware pause/resume buttons, persistent SD-card leaderboard.

FinFET Device Simulation – Semiconductor Modeling

Nov 2025 – Dec 2025

- Performed EDA-based FinFET simulation on nanoHUB's Synopsys TCAD (Sentaurus) for electrical characterization.
- Varied gate work function by 0.3 eV and interface-trap density by 10 $\times$, inducing V_T shifts up to 0.58 V, large I_{OFF} drop.
- Extracted V_T , SS, I_{ON} , I_{OFF} , and DIBL from I_D - V_G curves at $V_{DS} = 0.05$ V (linear) and 1.0 V (saturation) bias conditions.
- Correlated measured trends with manually calculated C_{ox} and C_{dep} models, achieving agreement within ~5% error.

Optical Heart Rate Sensor – Analog and Digital Design

Oct 2025 – Nov 2025

- Built infrared PPG sensor with IR204 LED & PT204 phototransistor, selecting resistors for stable mV-level waveform.
- Assembled cascaded high-pass and low-pass active filters with total gain ~250, isolating 40–200 BPM signals from noise.
- Created comparator with 3V hysteresis to generate -5 to 5 V pulses, driving indicator LED for real-time heartbeat detection.

Lunar Lander Game – *FPGA and Verilog*

Jul 2025 – Aug 2025

- Engineered modular Verilog components (ALU, memory, control, display units) for FPGA lunar lander simulation at 100 Hz.
- Devised 16-bit BCD arithmetic with adders/subtractors, enabling single-cycle updates for altitude, velocity, fuel, thrust.
- Verified implementation on hardware with 100+ test cases, driving 7-segment displays and LEDs for real-time user feedback.

Audio Equalizer Circuit – *Circuit Design*

Apr 2025 – May 2025

- Designed 3-band equalizer using op-amp RC/LC networks to implement low, mid, high-pass filters for frequency isolation.
- Combined amplification and potentiometer-based volume control, validating frequency response against LTSpice simulations.
- Reduced noise and distortion through proper biasing and filtering, allowing stable $\sim \pm 10$ dB per-band gain control.

Linux Shell Project – *Advanced Custom Bash Shell*

Jan 2024 – Mar 2024

- Constructed custom linux-like shell in C/C++ with Lex/Yacc, supporting subshells, loops, wildcards, pipes, I/O redirection.
- Configured modular process, environment manager with 10+ built-in commands & extensible scripting language features.
- Achieved broad feature coverage across 50+ detailed test cases, ensuring correctness of parsing, execution, and redirection.

Simple C Compiler – *Compiler Design*

Oct 2023 – Dec 2023

- Developed compiler from scratch for a subset of C using Lex and Yacc, translating source code into x86-64 assembly.
- Programmed stack-based expression evaluation, control-flow constructs, and function call handling with register optimization.
- Formulated symbol tables and scope management supporting global, local, and nested identifiers across 40+ test programs.

SeedTrace – *Android App Development*

May 2022 – Aug 2022

- Created SeedTrace Android app for supply chain transparency in organic food, giving 700+ consumers verifiable details.
- Programmed QR code scanning with ZXing in Java/JS to show farm origin, harvest date, warehouse location, certifications.
- Linked React Native frontend with REST APIs to deliver product data for 100+ items directly within mobile interface.

RESEARCH**Undergraduate Researcher** – *LLMs in Embedded Systems (RTOS)**Purdue University* | Jan 2026 – Present

- Led a 5-person team to benchmark LLM performance on embedded systems related PRs from Zephyr, NuttX & RIOT repos.
- Designed PR-sourcing pipeline that mined and filtered merged GitHub PRs into a curated, test-backed benchmark task set.
- Built Dockerized mini-swe-agent harness validating model-generated patches via tests in QEMU/Renode hardware sims.
- Evaluated five LLMs on pass-rate & step-efficiency against SWE-bench; all dropped on embedded tasks (best 70.6%).